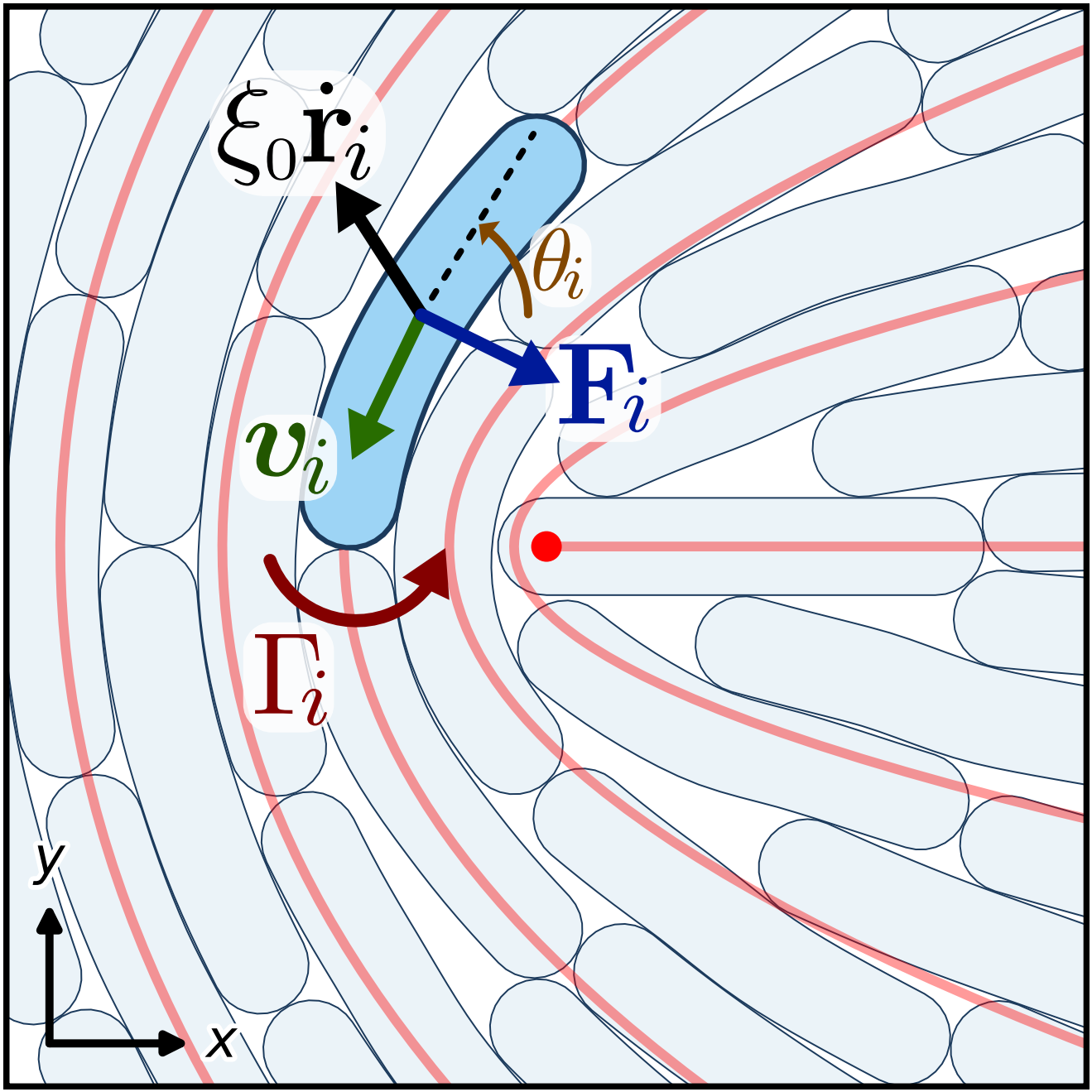
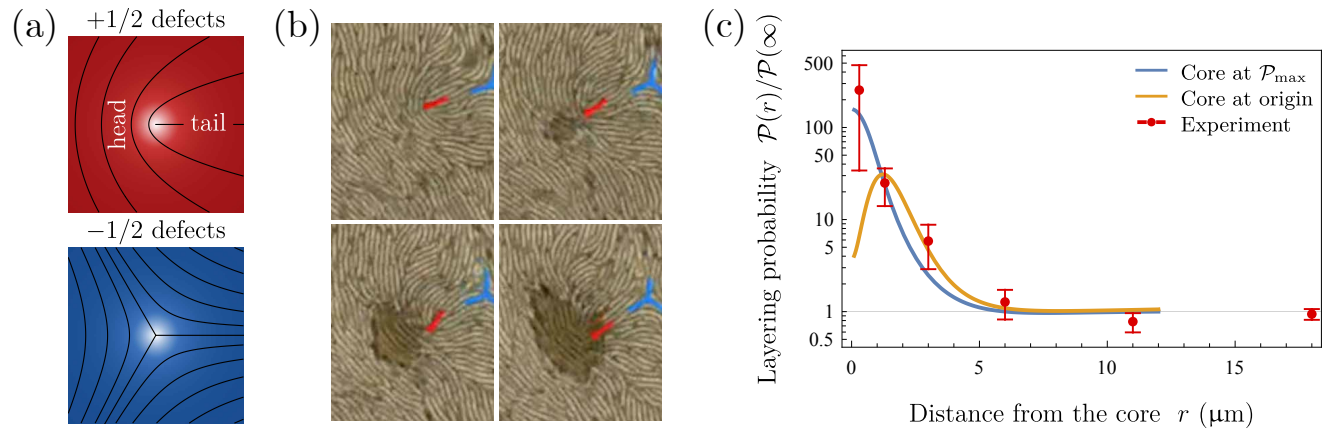
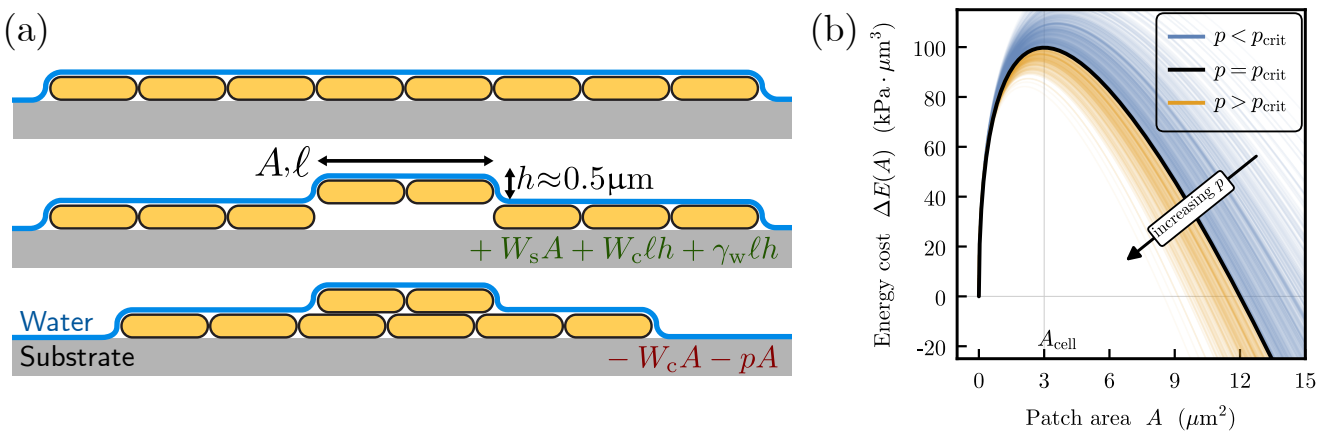

$\xi_0 \dot{\mathbf{r}}_i$
θ_i
$\mathbf{F}_i$
$\boldsymbol{v}_i$
Γ_i
y
x


(a)
+1/2 defects
head
tail
−1/2 defects
(b)
(c)
Layering probability $\mathcal{P}(r)/\mathcal{P}(\infty)$
500
100
50
10
5
1
0.5
0
5
10
15
Distance from the core r (μm)
Core at $\mathcal{P}_{\max}$
Core at origin
Experiment


(a)
A, ℓ
h ≈ 0.5 μm
+ W_s A + W_c ℓh + γ_w ℓh
Water
Substrate
− W_c A − pA
(b)
p < p_crit
p = p_crit
p > p_crit
increasing p
Energy cost ΔE(A) (kPa · μm³)
100
80
60
40
20
0
-20
A_cell
0
3
6
9
12
15
Patch area A (μm²)